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7 FEBRUARY 1975
(SEE SECTION 6)

MILITARY STANDARD

POWER CABLE ASSEMBLIES



FSC 6150

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DEPARTMENT OF DEFENSE
Washington, DC 20361

MIL-STD-1653(AS) POWER CABLE ASSEMBLIES

1. This Military Standard is approved for use by the Department of the Navy, Naval Air Systems Command.

2. Recommended corrections, additions, or deletions should be addressed to the Commanding Officer, Naval Air Engineering Center, ESSD (Code 93), Lakehurst, New Jersey 08733.

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1. GENERAL

1.1 Scope. This standard specifies the configuration requirements of input power cable assemblies for Organizational (flightline) and Intermediate (shop) electronic test equipment, classified as Type I in MIL-T-28800, and rated up to 10 amperes. It also shows the relevant connections for the test equipment and power source receptacles.

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

Federal

W-C-596	Connector, Electrical, Power

Military

MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-C-28777/5	Cable Assembly, Electronic Test Equipment, (3 wire, 3 prong, 125 volt, molded) Grounding Plug Connector
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for

STANDARDS

Military

MS2548	Cable Assemblies, Electric Power, for Portable Ground Support Equipment
MS3102	Connector, Receptacle, Electric, Box Mounting
MS3106	Connector, Plug, Electric, Straight
MS3107	Connector, Plug, Electric, Quick Disconnect
MS3436	Connector, Plug, Electric, Solder Contact, Utility

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STANDARDS (Continued)

Military

MS3507	Connector, Plug, Electric, Solder Contact, Quick Disconnect, Utility
MS90384	Cables, Electric Power, Permanently Wired, for Portable Ground Support Test Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as by the contracting officer.)

3. DEFINITIONS

3.1 Organization (flightline) equipment. Test equipment used in the vicinity of the aircraft on land or on board ship. It includes test equipment used on the flight line or flight deck, and in the hangar or hangar deck. This class of test equipment is for use in areas where the environmental conditions are uncontrolled.

3.2 Intermediate (shop) equipment. Test equipment used in the shop at land facilities or on board ship where the service area is protected and environmentally controlled for human occupancy. It includes test equipment used in field shops, fixed or mobile.

3.3 Type I. Test equipment designed specifically for military use.

4. GENERAL REQUIREMENTS

4.1 Electrical power source. Equipment not powered from aircraft or system under test shall operate from power sources specified by the procuring activity. The nominal power sources shall be selected from the following voltages that are readily available to the maintenance technician:

a. 60 and 400 Hz, single phase, 115V

b. 400 Hz (only), single phase, 115V

c. 400 Hz (only), 3 phase wye (line to neutral), 115/200V

d. DC, 28V

4.2 Cable assembly construction. This standard specifies the type of connectors, cables, color codes and pin connections required for each voltage source. Other items are needed to complete the cable assemblies to be consistent with good construction practices.

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5. DETAILED REQUIREMENTS

5.1 Connections.

a. The cable assembly connections for equipment rated up to 10 amperes shall be as specified in Table I, column 1 or 2 as applicable for the voltage source.

b. Connections for the source receptacles and equipment receptacles listed in Table I, columns 3 and 4 are shown for informational purposes.

c. Connections for other power source requirements or current ratings higher than 10 amperes shall be as specified by the procuring activity.

5.2 Power cable assemblies.

a. Detachable power cable assemblies shall be as specified in Table I, column 1 as applicable for the voltage source. They are furnished with equipment having a combination case enclosure with a removable front cover and input power connection on the front panel. The cables shall be stowed in the enclosure cover.

b. Permanently attached power cable assemblies shall be as specified in Table I, column 2 and shall be furnished with other style enclosures. A strain relief shall be provided.

c. Equipment operating from multiphase sources (60 and 400 HZ) shall be furnished with a cable assembly as specified in figures 1 or 11 and an adapter as in figure 15.

d. No branched power cable assemblies shall be furnished.

e. Power cable assemblies for other power sources or current ratings higher than 10 amperes shall be as specified by the procuring activity.

6. NOTES

6.1 This standard supercedes appendix A of MIL-T-21200L and appendix A of MIL-T-28800A.

Review activities:
Navy - EC

Preparing activity
Navy - AS

(Project No. 6150-N176)

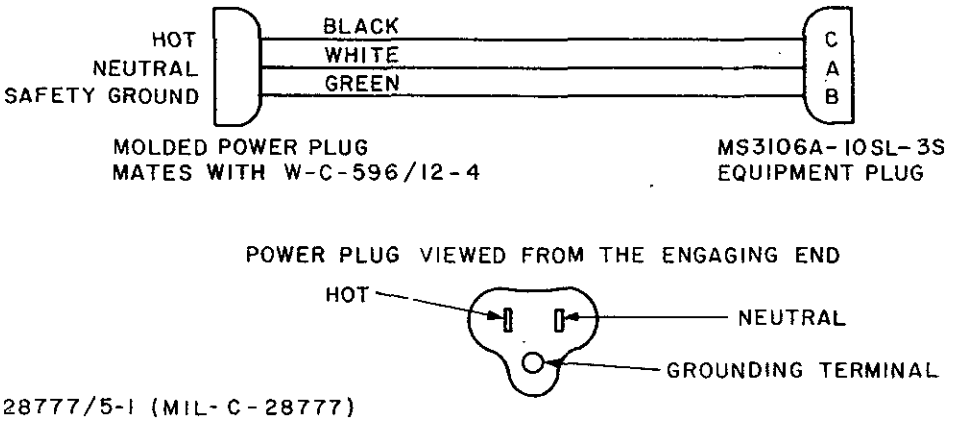
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TABLE I.

TEST EQUIPMENT POWER CABLES, CONNECTORS AND CONNECTIONS

	1	2	3	4
EQUIPMENT POWER REQUIREMENT	DETACHABLE POWER CABLE ASSEMBLY	ATTACHED POWER CABLE ASSEMBLY	EQUIPMENT RECEPTACLE	POWER SOURCE RECEPTACLE
60 and 400 HZ 1 Phase 115V	Figs. 1 & 15	Figs. 11 & 15	Fig. 2	Figs. 3 & 7
60 HZ 1 Phase 115V	Fig. 1	Fig. 11	Fig. 2	Fig. 3
400 HZ 1 Phase 115V	Fig. 4	Fig. 12	Fig. 5	Fig. 7
400 HZ 3 Phase Wye 115/200V	Fig. 4	Fig. 13	Fig. 6	Fig. 7
DC 28V	Fig. 5	Fig. 14	Fig. 9	Fig. 10

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CABLE NO: M28777/5-1 (MIL-C-28777)
LENGTH: 96 ± 2 INCHES
POWER CABLE P/N: MS2548-2

Figure 1. Power Cable (Detachable) 60 Hz 1φ 115 Volts.

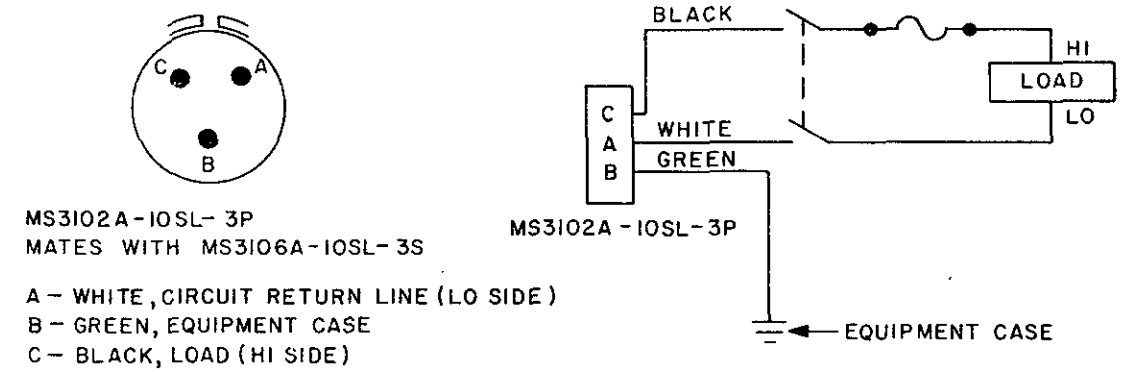
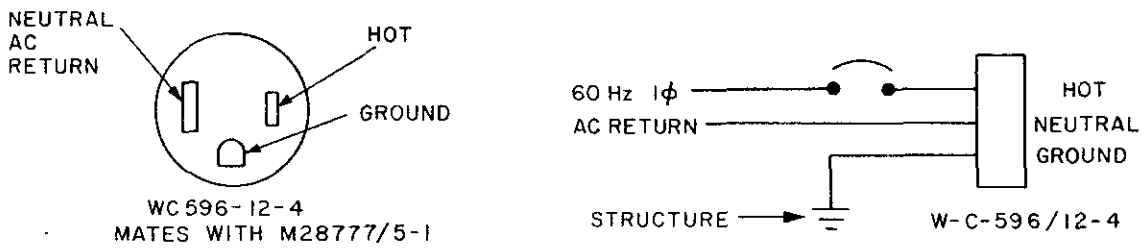


Figure 2. Test Equipment Receptacle 60 Hz 1φ 115 Volts.



HOT – 60 Hz 1φ
NEUTRAL – AC RETURN
GROUND – ATTACHED TO STRUCTURE

Figure 3. Bench Power Receptacle Wiring 60 Hz 1φ 115 Volts.

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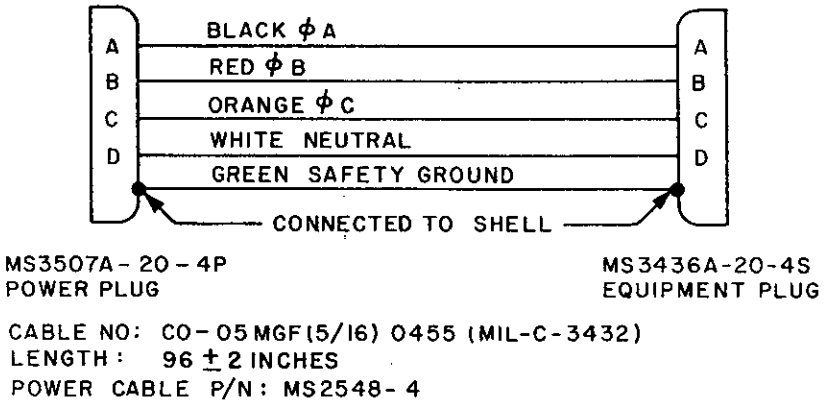


Figure 4. Power Cable (Detachable) 400 Hz 1ϕ and 3ϕ (line to neutral) 115/200V.

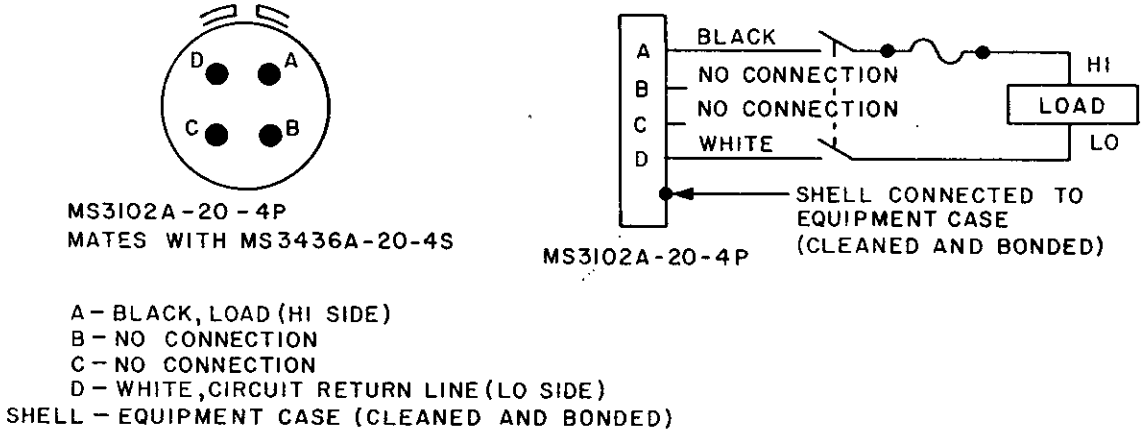
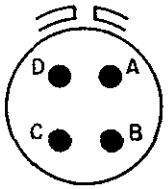
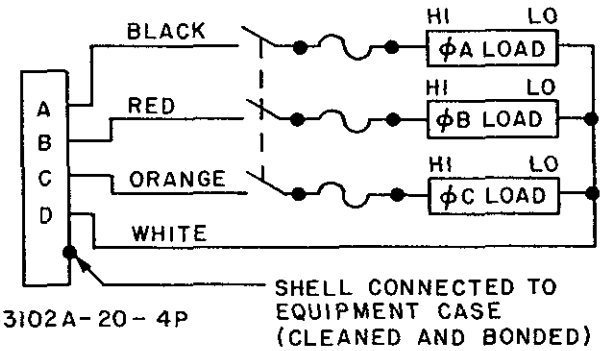


Figure 5. Test Equipment Receptacle 400 Hz 1ϕ 115 Volts.

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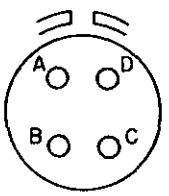


MS3102A-20-4P
MATES WITH MS3436A-20-4S

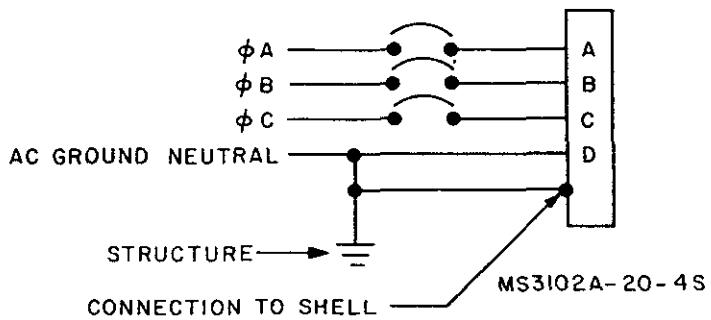


A - BLACK, ϕA LOAD (HI SIDE)
B - RED, ϕB LOAD (HI SIDE)
C - ORANGE, ϕC LOAD (HI SIDE)
D - WHITE, SYSTEM RETURN LINE (LO SIDE)
SHELL - EQUIPMENT CASE (CLEANED AND BONDED)

Figure 6. Test Equipment Receptacle 400 Hz 3ϕ (line to neutral) 115/200V.



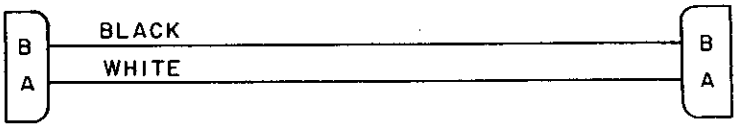
MS3102A-20-4S
MATES WITH MS3507A-20-4P



A - ϕA OR SINGLE ϕ
B - ϕB
C - ϕC
D - AC GROUND ATTACHED TO STRUCTURE
SHELL - GROUND ATTACHED TO STRUCTURE

Figure 7. Bench Power Receptacle Wiring 400 Hz 1ϕ and 3ϕ (line to neutral) 115/200V.

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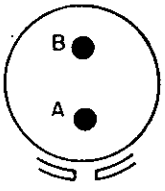


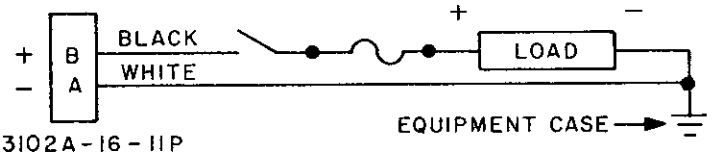
MS3107A-16-11P
POWER PLUG

MS3106A-16-11S
EQUIPMENT PLUG

CABLE NO. : CO-02MGF(2/14) 0400 (MIL-C-3432)
LENGTH : 96 ± 2 INCHES
POWER CABLE P/N: MS2548-5

Figure 8. Power Cable (Detachable) D.C. 28 Volts.



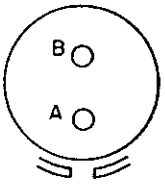


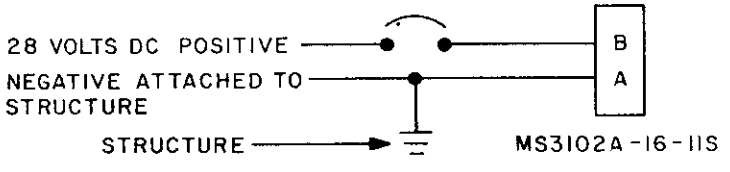
MS3102A-16-11P

MS3102A-16-11P
MATES WITH MS3106A-16-11S

A — WHITE, NEGATIVE GROUND
B — BLACK, POSITIVE, LOAD (HI SIDE)

Figure 9. Test Equipment Receptacle D.C. 28 Volts.





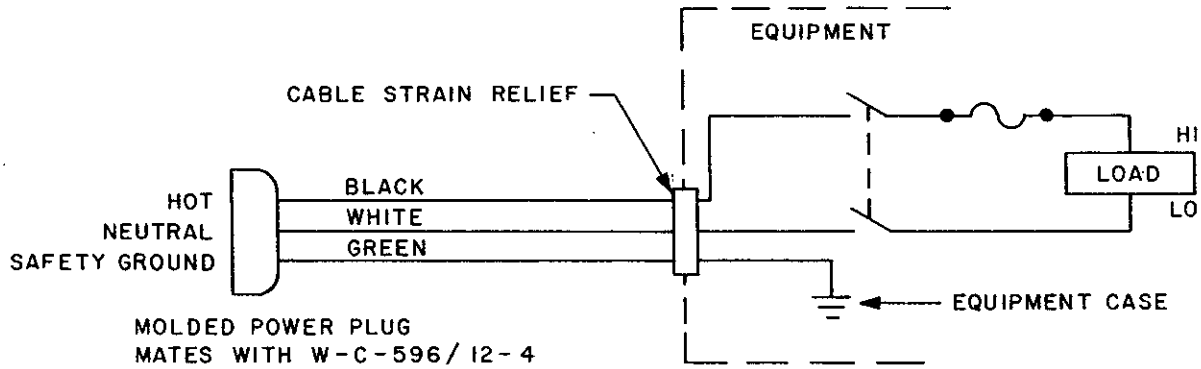
MS3102A-16-11S

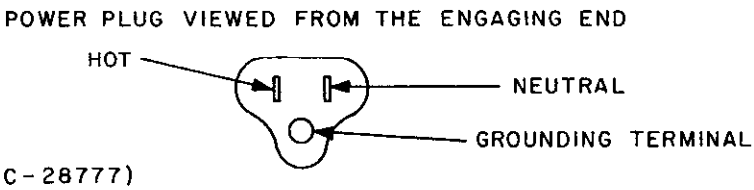
MS3102A-16-11S
MATES WITH MS3107A-16-11P

A — NEGATIVE, GROUND
B — POSITIVE, 28 VOLTS DC

Figure 10. Bench Power Receptacle Wiring D.C. 28 Volts.

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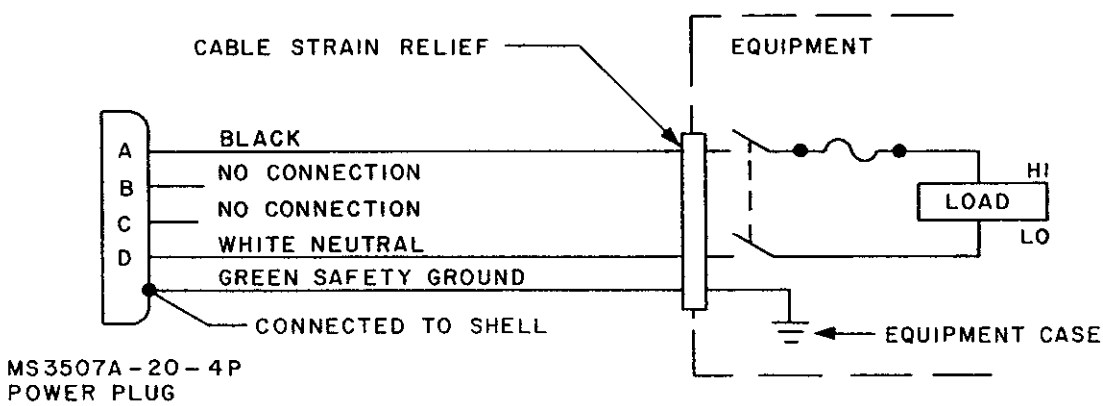


CABLE NO: M28777/5-1 (MIL-C-28777)

LENGTH: 96±2 INCHES, CABLE STRAIN RELIEF SHALL BE PROVIDED

POWER CABLE P/N: M28777/5-1

Figure 11. Power Cable (Attached) 60 Hz 1ϕ 115 Volts.



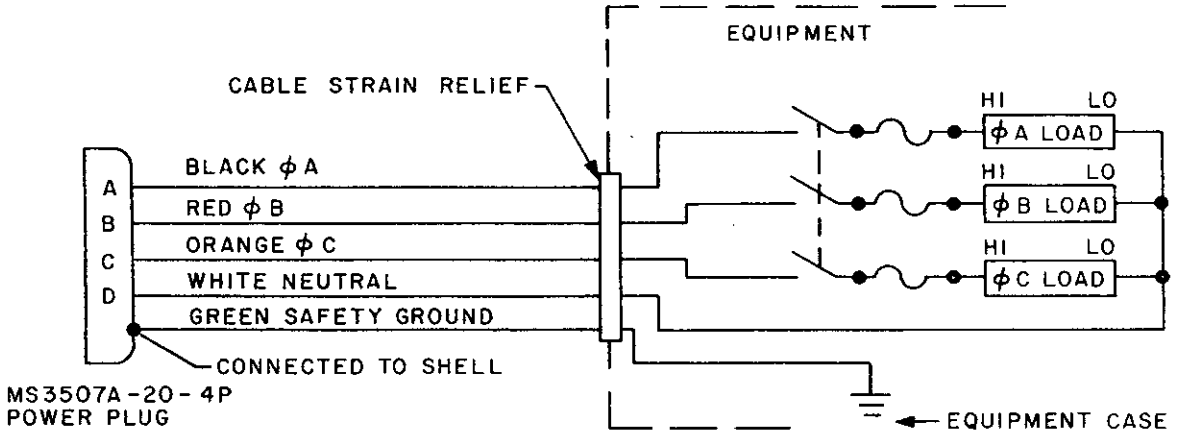
CABLE NO: CO-03MGF(3/16)0365 (MIL-C-3432)

LENGTH: 96±2 INCHES, CABLE STRAIN RELIEF SHALL BE PROVIDED

POWER CABLE P/N: MS90384-2

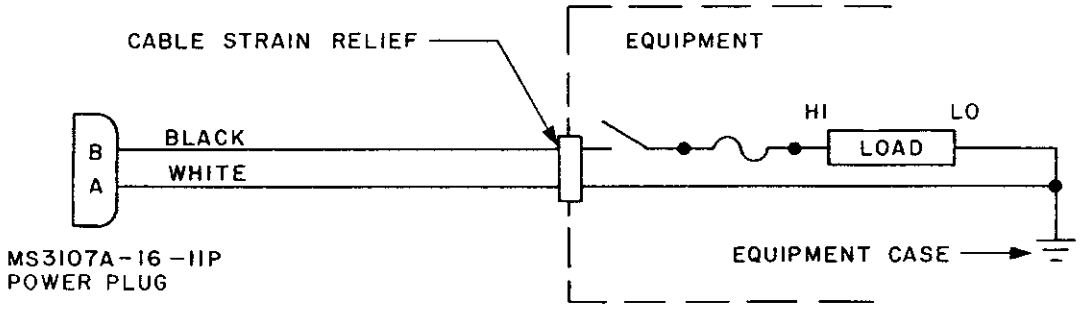
Figure 12. Power Cable (Attached) 400 Hz 1ϕ 115 Volts.

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CABLE NO: CO-05 MGF(5/16)0455(MIL-C-3432)
LENGTH: 96 ± 2 INCHES,CABLE STRAIN RELIEF SHALL BE PROVIDED
POWER CABLE P/N: MS90384-1

Figure 13. Power Cable (Attached) 400 Hz 3φ (line to neutral) 115/200 Volt.



CABLE NO: CO-02 MFG(2/14)0400(MIL-C-3432)
LENGTH: 96 ± 2 INCHES, CABLE STRAIN RELIEF SHALL BE PROVIDED
POWER CABLE P/N: MS90384-4

Figure 14. Power Cable (Attached) D.C. 28 Volts.

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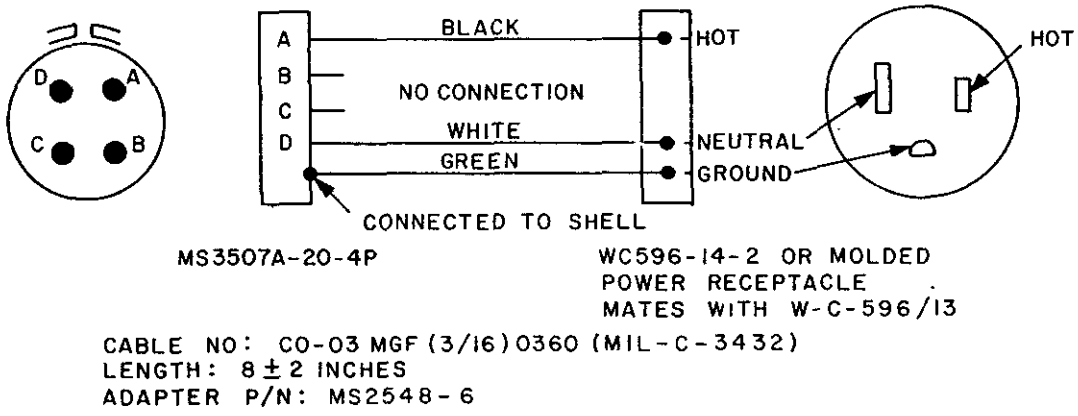


Figure 15. 60 Hz to 400 Hz Power Plug Adapter

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